



Midterm Exam I (90 minute exam)

Course: Electromagnetic Theory
Instructor: Dr. Shafayat Abrar

Date: March 15
Marks: 50 Marks

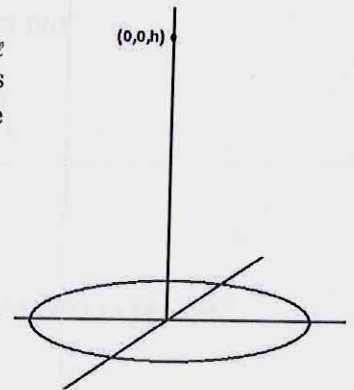
Instructions:

1. Attempt all questions.
2. A one-page formula-sheet is allowed.
3. Remember that talking is not allowed at any time in the exam hall.
4. Keep your eyes on your own paper. Remember, copying is cheating!
5. You are not permitted to share formula sheet, calculators or any other materials during the examination.
6. Remember! Cheating is not allowed and serious action will be taken if found guilty.
7. Stop writing immediately when the teacher says it is the end of the exam.

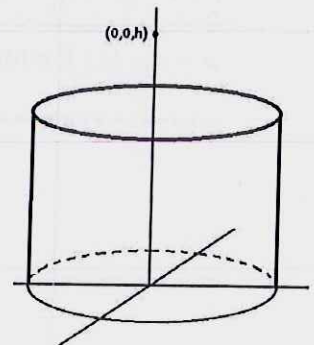
1. (08 points) A circular ring of radius a carries a uniform charge ρ_ℓ C/m and is placed on the xy -plane with axis the same as the z -axis. Providing all necessary steps, show that the electric potential at $(0, 0, h)$ is given by

$$V(0, 0, h) = \frac{kQ}{\sqrt{a^2 + h^2}}$$

where Q is the total charge on the ring. Also, provide the expression of Q .



2. (10 points) Providing all necessary steps, find the electric potential due to a hollow cylinder of charge (of radius a and height H) with uniform (surface) charge density ρ_S , at point $(0, 0, h)$ along the axis of symmetry (the z -axis).



Hint for Question 2 : Chop up the cylinder into thin rings at height z' , with thickness dz' . Each such ring has total charge $(\rho_s 2\pi a) dz'$, which can be inserted into the result obtained in Question 1, noting that the vertical distance to this ring is now $(h - z')$. Finally integrate over z' from 0 to H . You may require the following formula for integration:

$$\int \frac{dx}{\sqrt{b^2 + (x - a)^2}} = -\sinh^{-1} \left(\frac{x - a}{b} \right) + c$$

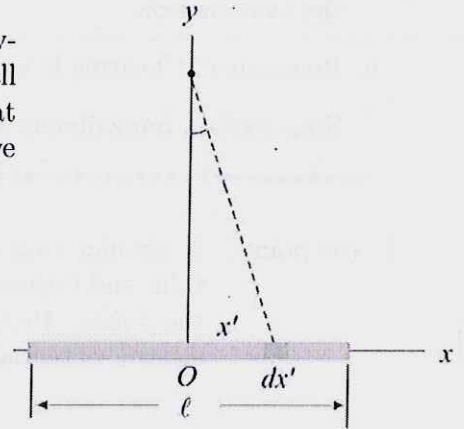
3. (08 points) An electric field in free space is $\mathbf{E} = (5z^2/\epsilon_0)\mathbf{a}_z$ V/m. Find the total charge contained within a cube, centered at the origin, of 4-m side length, in which all sides are parallel to coordinate axes (and therefore, each side intersects the axis at ± 2).

Hint : Try Gauss's law.

4. (10 points) Consider a non-conducting rod of length ℓ having a uniform charge density ρ_ℓ . Providing all necessary steps, find the electric potential at $(0, h)$, a perpendicular distance at y -axis above the midpoint of the rod.

You may require the following integration:

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left(x + \sqrt{x^2 + a^2} \right) + c$$



5. (14 points) Volume charge density is located as follows: $\rho_v = 0$ for $\rho < 1$ mm and for $\rho > 2$ mm; $\rho_v = 4\rho \mu\text{C}/\text{m}^3$ for $1 < \rho < 2$ mm. (a) Calculate the total charge in the region $0 < \rho < \rho_i$, $0 < z < L$, where $1 < \rho_i < 2$ mm. (b) Use Gauss's law to determine D_ρ at $\rho = \rho_i$. (c) Evaluate D_ρ at $\rho = 0.8$ mm, 1.6 mm, and 2.4 mm.
